Ps 07

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1 Problem 1 DONE

1.1 Task

Let $f : \mathbb{R} \to \mathbb{R}$ be a polynomial of degree $m \in \mathbb{N}$. Prove the following: (a) The Taylor polynomial of order $n \geq m$ located in $x_0 = 0$ coincides with f. (b) The Taylor polynomial of order $n \geq m$ located in $x_0 \in \mathbb{R}$ coincides with f.

1.2 Solution

- Part (a)

  Let the polynomial $f(x)$ be defined by its coefficients a_k:

$$f(x) = \sum_{k=0}^{m} a_k x^k$$

  The definition of the Taylor polynomial of order n at $x_0 = 0$ is:

$$T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k$$

  We evaluate the derivatives of f at 0:

  1. For $0 \leq k \leq m$: The k -th derivative of the term $a_k x^k$ is $a_k \cdot k!$. All other terms differentiate to 0 or vanish upon evaluation at $x = 0$. Thus, $f^{(k)}(0) = a_k \cdot k!$.
  2. For $k > m$: Since f has degree m, derivatives of order higher than m are identically zero. Thus, $f^{(k)}(0) = 0$.

  Substituting these into the summation (splitting at m):

$$\begin{aligned} T_n(x) &= \sum_{k=0}^{m} \frac{f^{(k)}(0)}{k!} x^k + \sum_{k=m+1}^{n} \frac{f^{(k)}(0)}{k!} x^k \\ &= \sum_{k=0}^{m} \frac{a_k \cdot k!}{k!} x^k + \sum_{k=m+1}^{n} \frac{0}{k!} x^k \\ &= \sum_{k=0}^{m} a_k x^k \\ &= f(x) \end{aligned}$$

- Part (b)

  We apply Taylor's Theorem with the Lagrange form of the remainder. For any $x \in \mathbb{R}$ and center $x_0 \in \mathbb{R}$:

$$f(x) = T_n(x) + R_n(x)$$

The remainder term is given by:

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0)^{n+1}$$

where ξ is some real number between x_0 and x.

Since f is a polynomial of degree m and we assumed $n \geq m$, it follows that $n + 1 > m$. The $(n + 1)$ -th derivative of a polynomial of degree m is the zero function:

$$f^{(n+1)}(t) = 0 \quad \text{for all } t \in \mathbb{R}$$

Therefore, $f^{(n+1)}(\xi) = 0$, which implies $R_n(x) = 0$.

$$\implies f(x) = T_n(x)$$

2 Problem 2 DONE

2.1 Task

Compute the Taylor polynomial of order $n \in \mathbb{N}$ at $x_0 = 0$ of

1. $f(x) = e^x$,
2. $f(x) = \sin(x)$,
3. $f(x) = \cos(x)$.

In addition, write Julia code that takes f and x_0 as input and returns the Taylor polynomial of order 2. Visualize that your above computations are correct when $n = 2$.

2.2 Solution

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + R_{n+1}(x)$$

1. $f(x) = e^x$

$$\begin{aligned}
T_n(x) &= \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k \qquad \text{(given } x_0 = 0\text{)} \\
f(0) &= e^0 = 1 \\
f^{(1)}(0) &= e^0 = 1 \\
f^{(n)}(0) &= e^0 = 1
\end{aligned}$$

Hence,

$$T_n(x) = \sum_{k=0}^{n} \frac{1}{k!} x^k = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + ... + \frac{1}{n!}x^n$$

2. $f(x) = \sin(x)$

$$
\begin{aligned}
t_n(x) &= \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k \qquad \text{(given } x_0 = 0\text{)} \\
f(0) &= sin(0) = 0 \\
f^{(1)}(0) &= cos(0) = 1 \\
f^{(2)}(0) &= -sin(0) = 0 \\
f^{(3)}(0) &= -cos(0) = -1 \\
&\text{hence,} \\
T_{2n+1}(x) &= \sum_{k=0}^{n} \frac{(-1)^k}{(2k+1)!} (x)^{2k+1} = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 + ...
\end{aligned}
$$

3. $f(x) = \cos(x)$

$$
\begin{aligned}
t_n(x) &= \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k \qquad \text{(given } x_0 = 0\text{)} \\
f(0) &= cos(0) = 1 \\
f^{(1)}(0) &= -sin(0) = 0 \\
f^{(2)}(0) &= -cos(0) = -1 \\
f^{(3)}(0) &= sin(0) = 0 \\
&\text{hence,} \\
T_{2n}(x) &= \sum_{k=0}^{n} \frac{(-1)^k}{(2k)!} (x)^{2k} = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - ...
\end{aligned}
$$

4. Code

```
using ForwardDiff
using Polynomials
using Plots

function get_taylor_polynomial(f, x_0 = 0.0, order = 2)
    coeff = zeros(Float64, order + 1)
    coeff[1] = f(x_0)

    if (order >= 1)
        coeff[2] = ForwardDiff.derivative(f, x_0)
    end
```

```
    if (order >= 2)
        f_prime = x -> ForwardDiff.derivative(f, x)
        second_deriv = ForwardDiff.derivative(f_prime, x_0)
        coeff[3] = second_deriv / factorial(2)
    end

    return Polynomial(coeff)
end

data = [
    (f = exp, label="exp(x)"),
    (f = sin, label="sin(x)"),
    (f = cos, label="cos(x)")
]

x_range = range(-2, 2, length=100)
p = plot(layout=(3,1), size=(600, 800), legend=:topleft)

for (i, item) in enumerate(data)
    poly_data = get_taylor_polynomial(item.f)
    plot!(p[i], x_range, item.f.(x_range), label="True $(item.label)", lw=3)
    plot!(p[i], x_range, poly_data.(x_range), label="Taylor Approx (n=2)", ls=:dash, lw=3
end

savefig("./figs/p_2.png")
```

```
/Users/ziro2173/personal/university/01_bachelor/2025_winter_term/01_analysis/02_exercises/s
```

3 Problem 3 DONE

3.1 Task

Determine those $s \in \mathbb{R}$, for which the integral

$$\int_0^1 x^s \, dx$$

is well-defined. In addition, illustrate your theoretical results by approximating the integral via the mid-point rule for a few distinct s in Julia.

3.2 Solution

Since the integrand x^s may have a singularity at $x = 0$, we treat this as an improper integral defined by the limit:

$$\int_0^1 x^s \, dx = \lim_{z \to 0^+} \int_z^1 x^s \, dx$$

We distinguish two main scenarios for the antiderivative: $s \neq -1$ and $s = -1$.

- Scenario A: $s \neq -1$ The antiderivative is power-based:

$$\begin{aligned}\int_z^1 x^s \, dx &= \left[\frac{x^{s+1}}{s+1}\right]_z^1 \\ &= \frac{1}{s+1} - \frac{z^{s+1}}{s+1}\end{aligned}$$

  We analyze the limit of the term $\frac{z^{s+1}}{s+1}$ as $z \to 0^+$:

  - Case 1: $s > -1 \implies s + 1 > 0$ The exponent is positive.

$$\lim_{z \to 0^+} z^{s+1} = 0$$

$$\implies \int_0^1 x^s \, dx = \frac{1}{s+1} - 0 = \frac{1}{s+1} \quad \text{(Converges)}$$

  - Case 2: $s < -1 \implies s + 1 < 0$ The exponent is negative. Let $p = -(s+1) > 0$.

$$\lim_{z \to 0^+} z^{s+1} = \lim_{z \to 0^+} \frac{1}{z^p} = \infty$$

$$\implies \int_0^1 x^s \, dx = \infty \quad \text{(Diverges)}$$

- Scenario B: $s = -1$ The power rule does not apply; the antiderivative is logarithmic.

$$\begin{aligned}\int_z^1 x^{-1} \, dx &= [\ln(x)]_z^1 \\ &= \ln(1) - \ln(z) \\ &= -\ln(z)\end{aligned}$$

  Taking the limit:

$$\lim_{z \to 0^+} (-\ln(z)) = -(-\infty) = \infty \quad \text{(Diverges)}$$

- Conclusion The integral is well-defined (convergent) if and only if:

$$s > -1$$

- Code

```
using Printf

function M(f, P)
    area = 0.0
    for i in 1:length(P)-1
        mid = (P[i+1] + P[i]) / 2
```

```
        h = f(mid)
        w = P[i+1] - P[i]
        area += h * w
    end
    return area
end

n_intervals = 10_000
s_values = [0.5, 0.0, -0.5, -0.9, -1.0, -1.5]

println("Approximating integral of x^s from 0 to 1 with $n_intervals intervals:\n")
@printf("%-10s | %-15s | %-15s | %-15s\n", "s", "Approx", "Exact", "Error")
println("-"^62)

for s in s_values

    f(x) = x^s

    P = range(0, 1, n_intervals)
    approx = M(f, P)

    if s > -1
        exact = 1 / (s + 1)
        error = abs(approx - exact)
        @printf("%-10.2f | %-15.5f | %-15.5f | %-15.5e\n", s, approx, exact, error)
    else
        exact_str = "Diverges"
        @printf("%-10.2f | %-15.5f | %-15s | %-15s\n", s, approx, exact_str, "N/A")
    end
end
```

```
M (generic function with 1 method)
10000
6-element Vector{Float64}:
  0.5
  0.0
 -0.5
 -0.9
 -1.0
 -1.5
Approximating integral of x^s from 0 to 1 with 10000 intervals:

s          | Approx          | Exact           | Error
--------------------------------------------------------------
0.50       | 0.66667         | 0.66667         | 6.06892e-08
0.00       | 1.00000         | 1.00000         | 0.00000e+00
```

```
-0.50      | 1.99395        | 2.00000        | 6.04929e-03
-0.90      | 6.74859        | 10.00000       | 3.25141e+00
-1.00      | 11.17375       | Diverges       | N/A
-1.50      | 475.62991      | Diverges       | N/A
```

4 Problem 4 DONE

4.1 Task

Let $p \in \mathbb{R}$ and consider the series

$$\sum_{n=2}^{\infty} \frac{1}{n(\log(n))^p}$$

Decide for which values of p the series converges.

4.2 Solution

We use the Integral Test. Let

$$f(x) = \frac{1}{x(\log(x))^p}$$

This function is positive and decreasing for $x \geq 2$.

We evaluate the improper integral:

$$\int_2^{\infty} \frac{1}{x(\log(x))^p}\, dx$$

Perform the substitution $u = \log(x)$, so $du = \frac{1}{x}dx$. The limits change: $x = 2 \implies u = \log(2)$ and $x \to \infty \implies u \to \infty$.

$$\int_2^{\infty} \frac{1}{x(\log(x))^p}\, dx = \int_{\log(2)}^{\infty} u^{-p}\, du$$

- Scenario A: $p \neq 1$ The antiderivative is $\frac{u^{1-p}}{1-p}$.

$$\int_{\log(2)}^{\infty} u^{-p}\, du = \left[\frac{u^{1-p}}{1-p}\right]_{\log(2)}^{\infty} = \frac{1}{1-p}\left(\lim_{u\to\infty} u^{1-p} - (\log(2))^{1-p}\right)$$

  - Case 1: $p < 1 \implies 1 - p > 0$ The exponent is positive, so $\lim_{u\to\infty} u^{1-p} = \infty$. The integral **diverges**.

- Case 2: $p > 1 \implies 1 - p < 0$ The exponent is negative, so $\lim_{u\to\infty} u^{1-p} = 0$. The integral **converges**.

- Scenario B: $p = 1$ The integral becomes:

$$\int_{\log(2)}^{\infty} \frac{1}{u}\, du = [\ln(u)]_{\log(2)}^{\infty} = \infty$$

  The integral **diverges**.

- Conclusion The series converges if and only if $p > 1$.